CONTINUOUS IMAGES OF QUASI-COMPACT SPACES ARE QUASI-COMPACT, AND QUASI-COMPACT MAPS HAVE RETROCOMPACT IMAGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of quasi-compactness for a space and for a map, and the notion of a retrocompact subset. Set up in the author's sections on quasi-compact spaces and maps, immediately before the counted statement. Fixes the two hypotheses and the two conclusions.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces.*

(1) *If X is quasi-compact, then $f(X)$ is quasi-compact.*
(2) *If f is quasi-compact, then $f(X)$ is retrocompact.*

Proof. If $f(X) = \bigcup V_i$ is an open covering, then $X = \bigcup f^{-1}(V_i)$ is an open covering. Hence if X is quasi-compact then $X = f^{-1}(V_{i_1}) \cup \ldots \cup f^{-1}(V_{i_n})$ for some $i_1, \ldots, i_n \in I$ and hence $f(X) = V_{i_1} \cup \ldots \cup V_{i_n}$. This proves (1). Assume f is quasi-compact, and let $V \subset Y$ be quasi-compact open. Then $f^{-1}(V)$ is quasi-compact, hence by (1) we see that $f(f^{-1}(V)) = f(X) \cap V$ is quasi-compact. Hence $f(X)$ is retrocompact. □

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